

SARAL SINGHAL

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Education

Vellore Institute of Technology

Bachelor's in Computer Science (B.Tech)

Bhopal, Madhya Pradesh

2023 – 2027

Maheshwari Public School

(Higher Secondary Education)

Jaipur, Rajasthan

2016 – 2022

Technical Skills

Languages: Java, Python, SQL

ML Frameworks: TensorFlow, Scikit-Learn, Keras

Data and Analytics: Pandas, Numpy, SHAP, Matplotlib

Concepts: Data Cleaning, EDA, Feature Engineering, Ensemble Learning

Tools: Git, GitHub, Google Colab, Jupyter Notebook, VS Code, Oracle BI

Soft Skills: Analytical Thinking, Effective Communicator, Time Management, Adaptability

Experience

Deloitte.

Intern - Strategy Risk Transactions

Gurugram, India

May 2026 – Present

- Working on developing research and analytical studies related to Public-Private Partnership (PPP) models and data centre infrastructure development in India.
- Conducting market research, feasibility analysis, and benchmarking on power, land, investment, and sustainability requirements for large-scale data centre projects.

Ernst & Young

Research Intern

Delhi, India

May 2025 – June 2025

- Market Analysis:** Performed a detailed assessment of non-fare box revenue streams for Chennai Metro Rail Limited (CMRL), evaluating commercial space utilization and land value capture mechanisms.
- System Specification:** Designed functional requirements and workflow models for a software platform to automate vendor-based non-fare revenue collection and monitoring.
- Technical Documentation:** Prepared and presented weekly technical reports and architectural briefs to senior consultants, communicating research insights and proposed system design.

Projects

ChurnSense Python, XGBoost, SMOTE, SHAP

[Link](#)

- Decision Intelligence Framework:** Designed a churn-risk and ROI optimization system quantifying customer lifetime value, retention costs, and campaign success probability to support data-driven business decisions.
- Strategic Customer Segmentation:** Translated predictive outputs into actionable retention strategies by segmenting customers into value-risk groups, enabling targeted marketing spend and intervention planning.
- Ensemble Modeling Pipeline:** Built a soft-voting ensemble combining Logistic Regression, Random Forest, and XGBoost with SMOTE balancing and standardized preprocessing for robust churn prediction.
- Interactive Analytics Platform:** Developed a Streamlit dashboard integrating profit curves, churn-risk scoring, K-Means clustering, and explainability modules for end-to-end churn management.

BleedDetectX Python, TensorFlow, Keras, PyTorch

[Link](#)

- Medical Imaging Models:** Developed deep learning architectures including U-Net, ResNet-UNet, and DMP-WCEBleedNet for bleeding classification and segmentation in Wireless Capsule Endoscopy images.
- High-Performance Results:** Achieved peak performance using DMP-WCEBleedNet with a Dice Coefficient of 0.812 and classification accuracy of 98.31% on the test dataset.
- Multi-Task Learning:** Built a PyTorch-based ResNet-UNet leveraging a pre-trained ResNet34 encoder for simultaneous bleeding classification and segmentation.
- Comprehensive Evaluation:** Deployed DMP-WCEBleedNet with multi-metric assessment including Dice, IoU, Precision, Recall, and F1-Score to ensure clinical-grade reliability.

Certifications

- Microsoft Certified: Data Engineer Associate
- Oracle Analytics Cloud Certified Professional
- Google Skill Badge: Inspect Rich Documents with Gemini Multimodality and Multimodal RAG (Vertex AI)
- Google Skill Badge: Prompt Design in Vertex AI
- Applied Machine Learning Specialization – Coursera